Coordinate-free Mathematics

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1 Algebra

1.1 Linear algebra

Fix a field k. A **k-vector space** is an abelian group $(V, +)$ with a scalar multiplication $k \times V \to V$ satisfying

$$\alpha(v+w) = \alpha v + \alpha w, \qquad (\alpha+\beta)v = \alpha v + \beta v,$$
$$(\alpha\beta)v = \alpha(\beta v), \qquad 1v = v$$

for $\alpha, \beta \in k$ and $v, w \in V$. A **linear map** $f : V \to U$ between k-vector spaces satisfies

$$f(\alpha v + \beta w) = \alpha f(v) + \beta f(w)$$

for $\alpha, \beta \in k$ and $v, w \in V$.

Exercise 1 (Small string vibrations obey superposition)
Derive the wave equation starting from a taut string and show that its solutions form a vector space. Show that time evolution from initial data to the state at time t is a linear map.

The **free k-vector space on a set** X, denoted kX, is the set of finitely supported set-theoretical functions $X \to k$ with pointwise addition and scalar multiplication. The **direct sum** vector space $V \oplus W$ is the set $V \times W$ with pointwise addition and scalar multiplication defined by $\alpha(v \oplus w) = \alpha v \oplus \alpha w$. The **quotient** of V by a subspace W, denoted V/W, is the set of equivalence classes of V modulo the relation $x \sim y$ iff $x - y \in W$. The **tensor product** $V \otimes W$ is $k(V \times W)/N$, where N is spanned by

$$(v+v', w) - (v, w) - (v', w), \quad (v, w+w') - (v, w) - (v, w'),$$
$$(\lambda v, w) - \lambda(v, w), \qquad (v, \lambda w) - \lambda(v, w)$$

for $v, v' \in V$, $w, w' \in W$, and $\lambda \in k$.

An **isomorphism** of vector spaces $V \cong W$ consists of linear maps $f : V \to W$ and $g : W \to V$ such that $f \circ g = 1_W$ and $g \circ f = 1_V$.

Exercise 2 (Universal constructions)
Give the universal properties of free spaces, direct sums, quotients, and tensor products, and use them to prove the three isomorphism theorems and $k(X \sqcup Y) \cong kX \oplus kY$, $k(X \times Y) \cong kX \otimes kY$.

The **dual space** V^* is the space of linear maps from a vector space V to its ground field k. The evaluation map $\mathrm{ev}_V : V^* \otimes V \to k$ is the linear extension of the evaluation morphism $f \otimes x \mapsto f(x)$. We say that V is **finite dimensional** if there is a co-evaluation morphism $\mathrm{coev}_V : k \to V \otimes V^*$ satisfying the snake equations $(1_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes 1_V) = 1_V$ and $(\mathrm{ev}_V \otimes 1_{V^*}) \circ (1_{V^*} \otimes \mathrm{coev}_V) = 1_{V^*}$. We define the **dimension** $\dim(V) = \mathrm{tr}(1_V)$, where $\mathrm{tr} : \mathrm{End}(V) \to k$ is given by $f \mapsto (\mathrm{ev}_V \circ s_{V,V^*} \circ (f \otimes 1_{V^*}) \circ \mathrm{coev}_V)(1)$ where $s_{X,Y} : X \otimes Y \to Y \otimes X$ is the symmetric braiding $x \otimes y \mapsto y \otimes x$.

Exercise 3 (Trace and dimension identities)
Prove the following for finite-dimensional vector spaces V, W.

- *If* $f : V \to W$ *and* $g : W \to V$*, then* $\mathrm{tr}(gf) = \mathrm{tr}(fg)$.
- *If* $f \in \mathrm{End}(V)$ *and* $g \in \mathrm{End}(W)$*, then* $\mathrm{tr}(f \oplus g) = \mathrm{tr}(f) + \mathrm{tr}(g)$.
- *If* $f \in \mathrm{End}(V)$ *and* $g \in \mathrm{End}(W)$*, then* $\mathrm{tr}(f \otimes g) = \mathrm{tr}(f)\,\mathrm{tr}(g)$.
- *If* $\mathrm{char}(k) = 0$*, then* $\dim(V) \in \mathbf{N} \subset k$
- *If* $\mathrm{char}(k) = 0$*, then* $V \cong W$ *iff* $\dim(V) = \dim(W)$

Define the tensor algebra and **exterior algebra of** V by

$$T(V) = \bigoplus_{k \geq 0} V^{\otimes k}, \qquad \Lambda V = T(V)/\langle v \otimes v : v \in V \rangle.$$

The space $\Lambda^k V$ is the degree-k component of ΛV.

Exercise 4 (The top exterior power is one-dimensional) *Let V be finite-dimensional with $n = \dim(V)$. Show that $\Lambda^n V$ is one-dimensional and that $\Lambda^k V = 0$ for all $k > n$.*

The **determinant** $\det(f)$ is the coefficient of the linear map on the highest-graded component of ΛV given by $v_1 \wedge \cdots \wedge v_{\dim(V)} \mapsto f(v_1) \wedge \cdots \wedge f(v_{\dim(V)})$.

Exercise 5 (Determinant identities and volume)
Prove the following:

- $\det(f \oplus g) = \det(f)\det(g)$
- *If* $V = \mathbb{R}^n$*, then* $\det(f)$ *is the signed volume of* $f(I^n)$ *where* $I = [0, 1]$.
- *Cramer's rule*

For $f \in \mathrm{End}(V)$, the **characteristic polynomial** is the polynomial $\lambda \mapsto \det(f - \lambda 1_V)$.

A scalar $\lambda \in k$ is an **eigenvalue** of f if $\ker(f - \lambda 1_V) \neq 0$. This kernel is its **eigenspace**; its dimension is the geometric multiplicity of λ.

Exercise 6 (Characteristic polynomial and Jordan form)
Let $f \in \mathrm{End}(V)$ have split characteristic polynomial. Prove the Cayley–Hamilton theorem and Jordan normal form, and derive the algebraic and geometric multiplicity inequality together with the eigenvalue formulas for $\mathrm{tr}(f)$ *and* $\det(f)$.

Exercise 7 (Critical damping is a Jordan block)
Use Jordan normal form to solve $m\ddot{x} + b\dot{x} + \kappa x = 0$ in all three damping regimes, and show that critical damping is exactly $b^2 = 4m\kappa$, where a nontrivial Jordan block produces $te^{-bt/(2m)}$.

A **Hermitian inner product** on a complex vector space V is a positive-definite sesquilinear form $\langle \cdot, \cdot \rangle$, conjugate-linear in the first variable and linear in the second. It induces the norm $\|v\| = \sqrt{\langle v, v\rangle}$. For a linear map $A : V \to W$ between finite-dimensional Hermitian spaces, an **adjoint** is a map $A^* : W \to V$ satisfying

$$\langle Av, w\rangle = \langle v, A^* w\rangle.$$

An endomorphism A is **self-adjoint** if $A = A^*$, **normal** if $A^*A = AA^*$, **positive** if $\langle v, Av\rangle \geq 0$ for every v, and **unitary** if $A^*A = AA^* = 1$. Equip $\mathrm{End}(V)$ with the induced operator norm.

Exercise 8 (Finite-dimensional spectral theorem)

Develop the finite-dimensional spectral theorem: prove existence of adjoints and positive square roots, and prove that every normal operator has an orthogonal spectral decomposition

$$A = \sum_{\lambda} \lambda p_\lambda, \qquad \sum_{\lambda} p_\lambda = 1.$$

Deduce the singular-value decomposition

$$T = U\Sigma V^*.$$

Exercise 9 (The spectral decomposition produces normal modes)

For a fixed-end chain of N equal masses and springs, set $q_0 = q_{N+1} = 0$ and define

$$(Kq)_j = \frac{\kappa}{m}(2q_j - q_{j-1} - q_{j+1}).$$

Find the spectral decomposition of K and use it to determine all normal modes of $\ddot{q} + Kq = 0$, their frequencies, and the conserved energy

$$\mathcal{E}(q, \dot{q}) = \frac{m}{2}\|\dot{q}\|^2 + \frac{m}{2}\langle q, Kq\rangle.$$

Exercise 10 (Fourier modes solve the fixed-end wave equation)

Solve the fixed-end wave equation from Exercise 1 by Fourier series, prove conservation of wave energy, and recover its normal-mode frequencies as the continuum limit of Exercise 9.

1.2 Group representations

An action of a group G on a set X assigns to each $g \in G$ a bijection $x \mapsto gx$ such that $1x = x$ and $(gh)x = g(hx)$. A **representation** of G on a vector space V is a homomorphism

$$\rho : G \longrightarrow \mathrm{GL}(V).$$

A subspace $W \subseteq V$ is G**-invariant** if $\rho(g)W \subseteq W$ for every $g \in G$. Restriction gives the **subrepresentation** W, and $\rho(g)(v + W) = \rho(g)v + W$ gives the **quotient representation** V/W. A nonzero representation is **irreducible** if its only invariant subspaces are 0 and V.

For representations V and W, the direct sum, tensor product, and dual representations are defined by

$$\begin{aligned} g(v,w) &= (gv, gw), \\ g(v \otimes w) &= gv \otimes gw, \\ (g\phi)(v) &= \phi(g^{-1}v), \qquad \phi \in V^*. \end{aligned}$$

An **intertwiner** $T : V \to W$ is a linear map satisfying $T\rho_V(g) = \rho_W(g)T$ for every g. Write $\mathrm{Hom}_G(V,W)$ for the space of intertwiners and

$$V^G = \{v \in V : \rho(g)v = v \text{ for every } g \in G\}$$

for the fixed-point space.

Exercise 11 (Averaging produces invariants)
Let G be finite and let V be a complex representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} \rho(g).$$

Prove that P_G is the projection of V onto V^G. Prove also that averaging any Hermitian inner product over G produces a G-invariant Hermitian inner product, so every finite-dimensional complex representation of G is equivalent to a unitary representation.

1.3 Maschke's theorem and Schur's lemma

Exercise 12 (Maschke's theorem)
Let $W \subseteq V$ be G-invariant and let $P : V \to W$ be any linear projection. Define

$$\widetilde{P} = \frac{1}{|G|} \sum_{g \in G} \rho(g) P \rho(g)^{-1}.$$

Prove that $\widetilde{P}$ is G-equivariant and has image W. Deduce that $\ker \widetilde{P}$ is a G-invariant complement of W. If G is finite and the characteristic of k does not divide $|G|$, prove that every finite-dimensional k-representation of G is a direct sum of irreducible representations.

Exercise 13 (Schur's lemma)
Let V and W be irreducible finite-dimensional complex representations. Prove that every nonzero intertwiner $T : V \to W$ is an isomorphism and that

$$\mathrm{End}_G(V) = \mathbb{C}1_V.$$

Let $\mathbb{C}[G]$ act on V by the linear extension of ρ. Deduce that every element of the center $Z(\mathbb{C}[G])$ acts by a scalar on each irreducible representation.

1.4 Characters and orthogonality

The **character** of a finite-dimensional representation V is the function

$$\chi_V(g) = \operatorname{tr}(\rho_V(g)).$$

For class functions $\chi, \psi : G \to \mathbb{C}$, define

$$\langle \chi, \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} \psi(g).$$

Exercise 14 (Great Orthogonality Theorem)
For irreducible unitary representations Γ^α and Γ^β of dimensions d_α, d_β of a finite group G, prove

$$\sum_{g \in G} \Gamma^\alpha(g)_{ij} \overline{\Gamma^\beta(g)_{k\ell}} = \frac{|G|}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{j\ell}.$$

Develop its character consequences, including

$$m_\alpha = \langle \chi_\alpha, \chi_V \rangle_G.$$

Deduce that characters classify finite-dimensional complex representations.

For an irreducible character χ_α, define the **symmetry projection**

$$P_\alpha = \frac{d_\alpha}{|G|} \sum_{g \in G} \overline{\chi_\alpha(g)} \rho(g).$$

Exercise 15 (Symmetry projections)
Prove that the operators P_α are pairwise orthogonal idempotents and that $P_\alpha V$ is the α-isotypic component of V. Interpret P_α as constructing symmetry-adapted linear combinations of atomic orbitals or displacement vectors.

1.5 Molecular symmetry

The **point group** of a molecule is the finite group $G \subseteq O(3)$ of spatial symmetries preserving its nuclear configuration, including atomic species. For a molecule with N atoms, G acts on the real displacement space

$$V_{\text{disp}} = \bigoplus_{a=1}^{N} \mathbb{R}^3_a \cong \mathbb{R}^{3N}$$

by permuting atoms and applying the spatial linear action to each displacement. The three-dimensional translational representation carries the polar-vector action R_g, while the rotational representation carries the axial-vector action $\det(R_g)R_g$; denote them by V_{trans} and V_{rot}. We complexify these real representations when applying character theory. For a nonlinear molecule, the vibrational representation is the class

$$[V_{\mathrm{vib}}] = [V_{\mathrm{disp}}] - [V_{\mathrm{trans}}] - [V_{\mathrm{rot}}]$$

in the representation ring.

Exercise 16 (Character of molecular displacements)
For $g \in G$, prove that

$$\chi_{\mathrm{disp}}(g) = \sum_{\substack{a \\ g(a)=a}} \mathrm{tr}(R_g),$$

where $R_g : \mathbb{R}^3 \to \mathbb{R}^3$ is the spatial action and the sum is over atoms fixed by g. Prove that $\dim V_{\mathrm{vib}} = 3N - 6$ for a nonlinear molecule, and use character orthogonality to express the multiplicity of every irreducible representation in V_{vib}.

For water, take the z-axis along the C_2 axis and the molecule in the yz-plane. Its point group is

$$C_{2v} = \{E, C_2, \sigma_{xz}, \sigma_{yz}\}.$$

Exercise 17 (Water has three symmetry-classified vibrations)
Derive the C_{2v} character table and use it to show that water's vibrational representation is

$$V_{\mathrm{vib}} \cong 2A_1 \oplus B_2.$$

Identify and construct the three normal modes.

1.6 Spectroscopic selection rules

Let V_i and V_f be the symmetry types of initial and final states. The electric-dipole operator transforms as the vector representation $V_\mu = V_{\mathrm{trans}}$. A transition amplitude

$$\langle \psi_f, \mu\psi_i \rangle$$

is the matrix coefficient used below.

Exercise 18 (Symmetry determines electric-dipole transitions)
Derive the electric-dipole selection rule

$$1 \subseteq V_f^* \otimes V_\mu \otimes V_i.$$

Apply it to show that all three fundamental vibrations of water are infrared active. Compare the mode assignments with the compiled gas-phase fundamentals

$$\nu_2 = 1594.59 \text{ cm}^{-1}, \qquad \nu_1 = 3656.65 \text{ cm}^{-1}, \qquad \nu_3 = 3755.79 \text{ cm}^{-1}$$

for the bend, symmetric stretch, and asymmetric stretch [35]. Reconcile the compilation's B_1 label for the asymmetric stretch with the B_2 label above by comparing axis conventions.

Exercise 19 (Water's vibrations are Raman active)

Let the polarizability tensor carry $\mathrm{Sym}^2(V_\mu)$*. Prove that a vibrational mode is Raman active precisely when its irreducible representation occurs in this symmetric square. Prove that*

$$\mathrm{Sym}^2(V_\mu) \cong 3A_1 \oplus A_2 \oplus B_1 \oplus B_2$$

for C_{2v}*. Deduce that all three fundamental vibrational modes of water are Raman active.*

1.7 Lie groups, infinitesimal symmetry, and spin

The molecular point groups above are discrete subgroups of $O(3)$. Define the group of orientation-preserving spatial rotations by

$$\mathrm{SO}(3) = \{R \in M_3(\mathbb{R}) : R^T R = 1,\ \det R = 1\}.$$

A **matrix Lie group** is a subgroup $G \subseteq \mathrm{GL}(n, \mathbb{C})$ that is closed in the subspace topology. Its **Lie algebra** is

$$\mathfrak{g} = \{X \in M_n(\mathbb{C}) : e^{tX} \in G \text{ for all } t \in \mathbb{R}\},$$

where $e^{tX} = \sum_{k \geq 0} (tX)^k/k!$. The Lie bracket is $[X, Y] = XY - YX$. An abstract **Lie algebra** over k is a k-vector space with a bilinear bracket that is skew-symmetric and satisfies the Jacobi identity. A **Lie algebra representation** on V is a linear map $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ with $\rho([X, Y]) = [\rho(X), \rho(Y)]$.

For $g \in G$ and $X, Y \in \mathfrak{g}$, the adjoint actions are

$$\mathrm{Ad}_g(X) = gXg^{-1}, \qquad \mathrm{ad}_X(Y) = [X, Y].$$

Exercise 20 (Classical matrix Lie algebras)

Prove that the Lie algebras of $\mathrm{GL}(n, \mathbb{C})$*,* $\mathrm{SL}(n, \mathbb{C})$*,* $\mathrm{O}(n)$*,* $\mathrm{SO}(n)$*,* $\mathrm{U}(n)$*, and* $\mathrm{SU}(n)$ *are respectively*

$$\mathfrak{gl}(n, \mathbb{C}), \quad \mathfrak{sl}(n, \mathbb{C}) = \{X : \mathrm{tr}(X) = 0\},$$

$$\mathfrak{o}(n) = \{X : X^T = -X\}, \quad \mathfrak{u}(n) = \{X : X^* = -X\}, \quad \mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}(n, \mathbb{C}).$$

Verify that each is a Lie algebra under $[X, Y] = XY - YX$*.*

Exercise 21 (One-parameter subgroups and the bracket)

Prove that $t \mapsto e^{tX}$ *is a smooth group homomorphism* $\mathbb{R} \to G$ *whenever* $X \in \mathfrak{g}$*, and that every smooth homomorphism* $\mathbb{R} \to G$ *arises uniquely in this way. Prove that*

$$[X, Y] = \left.\frac{d}{dt}\right|_{t=0} \left.\frac{d}{ds}\right|_{s=0} e^{tX} e^{sY} e^{-tX} e^{-sY},$$

and deduce $\mathrm{Ad}_{e^{tX}} = e^{t\,\mathrm{ad}_X}$*.*

Exercise 22 (Differentiating group representations)
Let $\rho : G \to \mathrm{GL}(V)$ be a smooth finite-dimensional representation of a matrix Lie group. Prove that

$$\dot{\rho}(X) = \frac{d}{dt}\bigg|_{t=0} \rho(e^{tX})$$

is a Lie algebra representation $\mathfrak{g} \to \mathfrak{gl}(V)$, and that $\rho(e^{tX}) = e^{t\dot{\rho}(X)}$. If G is connected, prove that a subspace is G-invariant if and only if it is $\mathfrak{g}$-invariant.

Exercise 23 (Rotations, angular momentum, and spin)
Using the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Identify $\mathfrak{su}(2) \cong \mathfrak{so}(3)$, construct the double cover $\mathrm{SU}(2) \to \mathrm{SO}(3)$, and classify the irreducible spin representations $\rho_j : SU(2) \to U(V_j)$. For $J_a = i\dot{\rho}_j(-i\sigma_a/2)$, derive $\dim V_j = 2j+1$ and $J^2 = j(j+1)1_{V_j}$, including the distinction between integer and half-integer spin.

A **quantum number** is an eigenvalue used to label a joint eigenspace of commuting observables.

Exercise 24 (Energy and symmetry produce quantum numbers)
Let H be a self-adjoint operator on a finite-dimensional Hermitian space V with a commuting $\mathrm{SU}(2)$ action. Derive the joint quantum-number decomposition

$$|E, j, m, \alpha\rangle$$

for H, J^2, J_3, including the allowed values and the degeneracy forced by spin j.

Exercise 25 (Hidden symmetry gives the hydrogen spectrum)
With $p = -i\hbar\nabla$, $r = |x|$, and $L = x \times p$, consider the hydrogen Hamiltonian and Runge–Lenz operator

$$H = \frac{p^2}{2\mu} - \frac{\kappa}{r}, \qquad A = \frac{1}{2\mu}(p \times L - L \times p) - \kappa\frac{x}{r}.$$

Use the hidden $\mathfrak{so}(4)$ symmetry to derive the bound-state spectrum and degeneracies

$$E_n = -\frac{\mu\kappa^2}{2\hbar^2 n^2}, \qquad n \geq 1, \qquad \dim W_n = n^2,$$

and the angular-momentum decomposition

$$W_n \cong \bigoplus_{\ell=0}^{n-1} V_\ell.$$

Recover the quantum numbers and transition frequencies, and compare them with hydrogen spectroscopy [34] and Pauli's algebraic derivation [33].

1.8 Monoidal and braided categories

A $\mathbb{C}$-linear category is a category whose morphism sets

$$\mathrm{Hom}_{\mathcal{C}}(x, y)$$

are complex vector spaces and whose composition is bilinear.

A $\mathbb{C}$-linear monoidal category is a $\mathbb{C}$-linear category equipped with a tensor product $\otimes$ that is bilinear on morphisms and is associative with tensor unit 1, up to coherent natural isomorphism.

Let $\mathcal{C}$ be such a category with finite direct sums. An object x is **simple** if $\mathrm{End}(x) \cong \mathbb{C}$, and $\mathcal{C}$ is **semisimple** if every object is a finite direct sum of simple objects. Assume that 1 is simple, that there are finitely many isomorphism classes of simple objects, and that $\mathcal{C}$ is **rigid**: every x has a dual x^* with evaluation and coevaluation maps satisfying the snake identities.

A **pivotal structure** on $\mathcal{C}$ is a monoidal natural isomorphism $j_x : x \to x^{**}$. It defines the right and left traces of $f : x \to x$ by

$$\begin{aligned}
\mathrm{tr}_R(f) &= \mathrm{ev}_{x^*}(j_x \otimes 1_{x^*})(f \otimes 1_{x^*})\,\mathrm{coev}_x, \\
\mathrm{tr}_L(f) &= \mathrm{ev}_x(1_{x^*} \otimes f)(1_{x^*} \otimes j_x^{-1})\,\mathrm{coev}_{x^*}\,.
\end{aligned}$$

The pivotal structure is **spherical** if $\mathrm{tr}_L(f) = \mathrm{tr}_R(f)$ for every endomorphism f. Assume that $\mathcal{C}$ has a spherical pivotal structure, write the common trace as $\mathrm{tr}(f)$, and set $d_x = \mathrm{tr}(1_x)$. Assume that the categorical dimensions of nonzero objects are positive real numbers.

Exercise 26 (Categorical trace and dimension)

Repeat the first three parts of Exercise 3 in $\mathcal{C}$.

Exercise 27 (The Lefschetz number is a categorical trace)

Show that bounded complexes of finite-dimensional vector spaces form a symmetric monoidal category and that the Lefschetz number defines a categorical trace.

For $x, y, z \in \mathcal{C}$, write

$$F^{xyz} : (x \otimes y) \otimes z \longrightarrow x \otimes (y \otimes z)$$

for the associator. A **braiding** is a family of isomorphisms

$$c_{x,y} : x \otimes y \longrightarrow y \otimes x.$$

For $f : x \to x'$ and $g : y \to y'$, it satisfies $(g \otimes f)c_{x,y} = c_{x',y'}(f \otimes g)$. The associator satisfies the pentagon identity, and the associator and braiding satisfy the hexagon identities. Suppressing associators, the latter are

$$\begin{aligned}
c_{x,y\otimes z} &= (1_y \otimes c_{x,z})(c_{x,y} \otimes 1_z), \\
c_{x\otimes y,z} &= (c_{x,z} \otimes 1_y)(1_x \otimes c_{y,z}).
\end{aligned}$$

The **monodromy** of x and y is the double braiding

$$M_{x,y} = c_{y,x}c_{x,y} : x \otimes y \longrightarrow x \otimes y.$$

Thus $c_{x,y}$ is one crossing, while $M_{x,y}$ is a full braid. If $x \otimes y$ is simple, identify $M_{x,y}$ with its scalar in $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Decompositions of tensor products of simple objects are **fusion rules**, and the spaces $\mathrm{Hom}_{\mathcal{C}}(z, x \otimes y)$ are **fusion spaces**.

An object x is **invertible** if $x \otimes x^* \cong 1$.

Exercise 28 (The $\nu = 1/3$ anyonic phase advances by $2\pi/3$)

Assume that every simple object is invertible and that their isomorphism classes form $\mathbb{Z}/m$, generated by a. Classify its monodromy, writing $M_{r,s} = M_{a^r,a^s}$, and show that, for some $k \bmod m$,

$$M_{r,s} = \exp\left(\frac{2\pi i k r s}{m}\right).$$

For $k = 1$ and $m = 3$, derive the $2\pi/3$ phase advance per enclosed anyon and compare it with [50].

Exercise 29 (Reshetikhin–Turaev TQFT)

A ribbon fusion category is a braided fusion category with a dual-compatible twist θ satisfying $\theta_{x\otimes y} = M_{x,y}(\theta_x \otimes \theta_y)$. For simple objects x_i, define $S_{ij} = \mathrm{tr}(M_{x_i,x_j})$ and call the category modular when S is invertible. Prove that every modular ribbon fusion category $\mathcal{C}$ defines a symmetric monoidal functor [44]

$$Z_{\mathcal{C}} : \widehat{\mathrm{Bord}}_3^{\mathcal{C}} \longrightarrow \mathrm{Vect}_{\mathbb{C}},$$

where $\widehat{\mathrm{Bord}}_3^{\mathcal{C}}$ is the anomaly-corrected category of oriented three-dimensional bordisms with $\mathcal{C}$-colored ribbon graphs.

2 Geometry

2.1 Differential forms and cohomology

A **smooth n-manifold** is a Hausdorff, second-countable topological space equipped with a maximal atlas of charts to open subsets of $\mathbb{R}^n$ whose transition maps are smooth. A smooth map $f : M \to N$ has a tangent map $df : TM \to TN$. The **tangent bundle** TM is the union of the tangent spaces, the **cotangent bundle** is T^*M, and a **vector field** is a smooth section of TM.

For an n-dimensional real vector space V, its **determinant line** is $\det(V^*) = \Lambda^n V^*$. An **orientation** of V is a choice of one of the two connected components of $\det(V^*) \setminus \{0\}$; the elements of the chosen component are positive. An orientation of an n-manifold M is a smoothly varying choice of orientation of its tangent spaces.

The exterior powers $\Lambda^k T_p^* M$ are those defined in the Algebra section. The space of **differential n-forms** is $\Omega^n(M) = \Gamma(\Lambda^n T^* M)$. On an oriented n-manifold, a **volume form** is a nowhere-vanishing element of $\Omega^n(M)$ that is positive with respect to the orientation. The exterior derivative $d : \Omega^n(M) \to \Omega^{n+1}(M)$ extends the differential of smooth functions and satisfies

$$d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$$

for $\alpha \in \Omega^p(M)$. A **singular n-simplex** is a smooth map $\sigma : \Delta^n \to M$ where Δ^n is the standard n-simplex. The collection of singular n-simplices in M freely generates the abelian group $C_n(M)$ of **singular n-chains** in M. The **boundary map** $\partial : C_n(M) \to C_{n-1}(M)$ is defined on each simplex $\sigma : \Delta^n \to M$ by

$$\partial\sigma = \sum_{j=0}^{n} (-1)^j \sigma \circ f_j,$$

where $f_j : \Delta^{n-1} \to \Delta^n$ is the inclusion of the face opposite the j-th vertex of Δ^n.

A **chain complex** $(A_n, d_n)_n$ is a sequence of abelian groups A_n and homomorphisms $d_n : A_n \to A_{n-1}$ such that $d_{n-1} \circ d_n = 0$ for all n. The n-th **homology group** H_n of the chain complex is

$$H_n = \ker(d_n)/\operatorname{im}(d_{n+1}).$$

The **singular homology** $H_n(M)$ is the homology of $(C_\bullet(M), \partial)$.

A **cochain complex** $(A^n, d^n)_n$ is a sequence of abelian groups and homomorphisms $d^n : A^n \to A^{n+1}$ satisfying $d^{n+1} d^n = 0$. Its **nth cohomology group** is

$$H^n(A^\bullet) = \ker d^n / \operatorname{im} d^{n-1}.$$

For an abelian group G, the **singular cochains** are

$$C^n(M; G) = \operatorname{Hom}(C_n(M), G), \qquad (d\varphi)(c) = \varphi(\partial c),$$

and their cohomology is the **singular cohomology** $H^n(M; G)$.

The **de Rham cohomology** of M is

$$H^n_{\mathrm{dR}}(M) = \ker(d\colon \Omega^n(M) \to \Omega^{n+1}(M))/\operatorname{im}(d\colon \Omega^{n-1}(M) \to \Omega^n(M)).$$

For a smooth map $f : N \to M$, the **pullback** of $\alpha \in \Omega^p(M)$ is

$$(f^*\alpha)_x(v_1, \ldots, v_p) = \alpha_{f(x)}(df_x v_1, \ldots, df_x v_p).$$

Exercise 30 (de Rham theorem)

Prove that integration over smooth singular cycles induces a natural isomorphism between de Rham cohomology and singular cohomology with real coefficients.

Exercise 31 (Cohomology produces the Aharonov–Bohm phase)
Outside an idealized solenoid, let

$$M = \mathbb{R}^3 \setminus \{(0,0,z) : z \in \mathbb{R}\}, \qquad d\theta = \frac{-y\,dx + x\,dy}{x^2+y^2}, \qquad A = \frac{\Phi}{2\pi} d\theta.$$

For a particle of charge q, use de Rham cohomology to derive the gauge-invariant phase

$$\exp\left(\frac{iq}{\hbar}\int_\gamma A\right)$$

and the Aharonov–Bohm interference shift $q\Phi/\hbar$ [47]. Explain how the shielded-flux electron-holography experiment of Tonomura et al. isolates this holonomy, and compare its measured relative phase shift with the prediction [48].

2.2 Hodge theory and electromagnetism

A **pseudo-Riemannian metric** is a smooth nondegenerate symmetric section $g \in \Gamma(T^*M \otimes T^*M)$. On an oriented pseudo-Riemannian n-manifold, the metric induces a bilinear form $\langle \cdot, \cdot \rangle_g$ on differential forms. A **metric volume form** vol_g is a positive volume form satisfying

$$|\langle \mathrm{vol}_g, \mathrm{vol}_g \rangle_g| = 1.$$

The **Hodge star** is an operator $\star : \Omega^p(M) \to \Omega^{n-p}(M)$ satisfying

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle_g \, \mathrm{vol}_g\,.$$

Exercise 32 (Metric volume and Hodge star)
Prove that an oriented pseudo-Riemannian manifold has a unique metric volume form and a unique Hodge-star operator. Determine $\star^2$ in terms of the degree and signature of the metric.

On an oriented Lorentzian four-manifold, the electromagnetic field strength is a two-form F and the electric four-current is a three-form J.

Exercise 33 (Maxwell's equations imply charge conservation)
From Maxwell's equations

$$dF = 0, \qquad d \star F = J.$$

Derive charge conservation in differential and integral form. On Minkowski spacetime, write

$$F = B - dt \wedge E, \qquad J = \rho\ \mathrm{vol}_3 - dt \wedge \star_3 j,$$

and recover the classical Maxwell equations and continuity equation.

Exercise 34 (Gauge invariance forces charge conservation)
For $F = dA$, consider

$$S[A] = -\frac{1}{2}\int_M F \wedge \star F + \int_M A \wedge J.$$

Derive Maxwell's equations and prove that gauge invariance is equivalent to charge conservation, assuming vanishing boundary terms.

2.3 Vector bundles and connections

A **smooth vector bundle** $p : E \to M$ is a smooth family of vector spaces locally isomorphic to a product $U \times V \to U$. For a smooth map $f : N \to M$, its **pullback bundle** is

$$f^*E = \{(x, e) \in N \times E : f(x) = p(e)\} \longrightarrow N.$$

Write $\Omega^0(M, E) = \Gamma(E)$. A **connection** on E is a ground-field linear map

$$\nabla : \Omega^0(M, E) \longrightarrow \Omega^1(M, E)$$

satisfying $\nabla(fs) = df \otimes s + f\nabla s$. Extend it to E-valued forms by, for $\alpha \in \Omega^p(M)$,

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^p \alpha \wedge \nabla\eta,$$

and define its **curvature** by $F_\nabla = \nabla^2 \in \Omega^2(M, \operatorname{End} E)$. Equip $\operatorname{End} E$-valued forms with the induced connection.

Exercise 35 (Bianchi identity)
Prove that every vector-bundle connection satisfies $\nabla F_\nabla = 0$.

2.4 K-theory

For a compact Hausdorff space X, let $\mathrm{Vect}(X)$ be the category of finite-rank complex vector bundles with continuous local trivializations over X, equipped with direct sum, tensor product, dual, and pullback. The **Grothendieck group** $K^0(X)$ is generated by isomorphism classes $[E]$ subject to

$$[E \oplus F] = [E] + [F].$$

Define multiplication by $[E][F] = [E \otimes F]$. For a continuous map $f : X \to Y$ between compact Hausdorff spaces, define pullback by

$$f^* : K^0(Y) \longrightarrow K^0(X), \qquad [E] \longmapsto [f^*E].$$

Exercise 36 (Grothendieck group of vector bundles)
Prove that $(f \circ g)^*E \cong g^*f^*E$ *and that pullback preserves direct sums, tensor products, and duals. Let* X *and* Y *be compact Hausdorff spaces and let* $f : X \to Y$ *be continuous. Prove that direct sum and tensor product induce a commutative ring structure on* $K^0(X)$ *and that* f^* *is a ring homomorphism. Prove that*

$$K^0(\mathrm{pt}) \cong \mathbb{Z}$$

via the rank map.

Fix a basepoint $x_0 \in X$, regarded as a map $x_0 : \mathrm{pt} \to X$. The **reduced K-theory** of the pointed space (X, x_0) is

$$\widetilde{K}^0(X) = \ker\left(K^0(X) \xrightarrow{x_0^*} K^0(\mathrm{pt})\right).$$

For a bundle E of rank r, its **reduced class** is

$$[E] - [\underline{\mathbb{C}}^r],$$

where $\underline{\mathbb{C}}^r = X \times \mathbb{C}^r$ is the trivial rank-r bundle. Write $K^n(X)$ for the generalized cohomology groups extending $K^0(X)$.

Exercise 37 (Stable equivalence and Bott periodicity)
Let E and F be complex vector bundles over a compact Hausdorff space X. Prove that $[E] = [F]$ in $K^0(X)$ if and only if there is a vector bundle G such that

$$E \oplus G \cong F \oplus G.$$

Deduce that the reduced class of E is unchanged after replacing E by $E \oplus \underline{\mathbb{C}}^n$. Prove Bott periodicity

$$K^{n+2}(X) \cong K^n(X)$$

and deduce that complex K-theory is naturally $\mathbb{Z}/2$-graded.

For $0 \leq r \leq N$, identify the complex Grassmannian $\mathrm{Gr}_r(\mathbb{C}^N)$ with the space of rank-r orthogonal projections in $M_N(\mathbb{C})$. Its **tautological bundle** has fiber $\operatorname{im} P$ over P. Let BU denote the stable complex Grassmannian obtained from finite Grassmannians by adjoining trivial summands.

Exercise 38 (Stable classification of complex vector bundles)
Classify stable complex vector bundles over a connected finite CW complex X by projection-valued maps, proving

$$K^0(X) \cong [X, \mathbb{Z} \times BU], \qquad \widetilde{K}^0(X) \cong [X, BU]_*.$$

Exercise 39 (Künneth theorem computes torus K-theory)
For finite CW complexes X and Y with torsion-free K-groups, prove that external product induces the graded isomorphism [1]

$$K^*(X) \mathbin{\widehat{\otimes}} K^*(Y) \cong K^*(X \times Y).$$

Here $\widehat{\otimes}$ denotes graded tensor product. Use Bott periodicity to deduce

$$K^*(\mathbb{T}^d) \cong \Lambda_{\mathbb{Z}}(\eta_1, \ldots, \eta_d), \qquad |\eta_j| = 1.$$

2.5 Chern classes and the Chern character

Let P be an invariant symmetric k-linear polynomial on the endomorphism algebra of a fiber of E, where invariant means invariant under simultaneous conjugation. Define

$$P(F_\nabla) = P(F_\nabla, \ldots, F_\nabla) \in \Omega^{2k}(M).$$

Exercise 40 (Chern–Weil theory produces Chern classes)

Prove that $P(F_\nabla)$ is closed and that its de Rham class is independent of the connection. Apply this to

$$\det\left(1+\frac{iF_\nabla}{2\pi}\right)=1+c_1(E,\nabla)+c_2(E,\nabla)+\cdots$$

to define the Chern classes of E [39].

Exercise 41 (Chern character)

Define

$$\mathrm{ch}(E,\nabla)=\operatorname{tr}\exp\left(\frac{iF_\nabla}{2\pi}\right).$$

For compact M, prove that it induces a ring homomorphism

$$\mathrm{ch}:K^0(M)\longrightarrow H^{\mathrm{even}}_{\mathrm{dR}}(M).$$

2.6 Band topology

For the gapped families below, assume that X is compact and connected, so spectra are uniformly bounded and vector-bundle ranks are constant. Let V be a finite-dimensional Hermitian vector space. Exercise 8 supplies the spectral decomposition used below. A continuous family is a map

$$H:X\longrightarrow \mathrm{End}(V)$$

and write $H_x=H(x)$. It is **gapped** if every H_x is self-adjoint and invertible. Given a positively oriented contour Γ enclosing the negative spectra and excluding the positive spectra, define

$$P_x^-=\frac{1}{2\pi i}\int_\Gamma (z-H_x)^{-1}\,dz,\qquad E_x^-=\mathrm{im}(P_x^-).$$

Exercise 42 (Gapped families define stable K-theory classes)

Show that the negative spectral subspaces of a gapped family form a vector bundle and that its class in $K^0(X)$ is invariant under gap-preserving homotopy. Show that

$$[E_H^-]-[\underline{\mathbb{C}}^{\,\operatorname{rank}E_H^-}]\in\widetilde{K}^0(X)$$

is invariant under adjoining constant gapped summands, and that every reduced class is represented by a flattened family $H_x=1-2P_x$.

Exercise 43 (Berry phase as holonomy)

Let $H:X\to\mathrm{End}(V)$ be a smooth self-adjoint family with a simple isolated eigenvalue λ_x, and let $L\to X$ be its eigenline bundle. For a local unit eigenvector ψ, define

$$\mathcal{A}=i\langle\psi,d\psi\rangle,\qquad \mathcal{F}=d\mathcal{A}.$$

Prove that these local forms define a connection and curvature on L and that the Berry phase around a closed curve γ is its holonomy $\exp(i\int_\gamma \mathcal{A})$. *For* $H_n = n\cdot\sigma$, $n \in S^2$, *prove that the phases of the two eigenlines around a loop enclosing oriented solid angle* Ω *are* $\gamma_\pm = \mp\Omega/2$ *modulo* 2π, *and compare this prediction with neutron spin rotation [49].*

Exercise 44 (K-theory classifies class A band insulators)

A class A band model over a compact connected parameter space X has a smooth gapped Hamiltonian

$$H : X \longrightarrow \mathrm{End}(V)$$

whose negative-energy bundle consists of occupied states; for a crystal, $X = \mathbb{T}^d$. *Prove that stable equivalence classes are classified by* $\widetilde{K}^0(X)$ *[4], and compute the strong phases*

$$\widetilde{K}^0(S^d) \cong \begin{cases} \mathbb{Z}, & d \textit{ even}, \\ 0, & d \textit{ odd}. \end{cases}$$

Use Exercise 39 to compute the classifications over $\mathbb{T}^d$ *for* $d = 1,2,3$ *and distinguish strong from weak invariants. For* $d = 2$, *use Exercise 40 to identify the integer invariant with the first Chern number*

$$C_1(E_H^-) = \frac{i}{2\pi}\int_{\mathbb{T}^2} \mathrm{tr}\big(P^-\, dP^- \wedge dP^-\big),$$

and prove that a nonzero value obstructs a global smooth frame of occupied states. Assuming the TKNN formula $\sigma_{xy} = C_1(E_H^-)e^2/h$ *[6], deduce the quantized Hall resistance and compare its plateaus and geometry independence with the original observation [5].*

2.7 Symplectic geometry

For a vector field X, contraction is the map $\iota_X : \Omega^p(M) \to \Omega^{p-1}(M)$ defined by

$$(\iota_X\alpha)(X_2,\dots,X_p) = \alpha(X, X_2,\dots,X_p).$$

Define the Lie derivative of forms by Cartan's formula $\mathcal{L}_X = d\iota_X + \iota_X d$.

A **symplectic form** is a two-form $\omega \in \Omega^2(M)$ such that $d\omega = 0$ and $X \mapsto \iota_X\omega$ is a bundle isomorphism $TM \to T^*M$. Write $C^\infty(M)$ for the smooth real-valued functions on M. For $H \in C^\infty(M)$, define X_H by

$$\iota_{X_H}\omega = dH.$$

Let $\pi : T^*Q \to Q$ be the cotangent bundle of a configuration manifold. Its **tautological one-form** is

$$\theta_{(q,p)}(\xi) = p\big(d\pi_{(q,p)}\xi\big),$$

and its **canonical symplectic form** is $\omega_0 = -d\theta$.

Exercise 45 (The symplectic form produces Hamilton's equations)
For

$$H(q,p) = \frac{1}{2m}\sum_{i=1}^{n} p_i^2 + V(q),$$

derive Hamilton's equations intrinsically from ω_0, recover Newton's equation, and solve the harmonic-oscillator flow.

Exercise 46 (Magnetic curvature produces the Lorentz force)
For a particle of charge e in a closed magnetic field $B \in \Omega^2(\mathbb{R}^3)$, derive the Lorentz force from

$$\omega_B = \omega_0 - e\pi^* B \qquad \textit{and} \qquad H(q,p) = \frac{|p|^2}{2m} + e\phi(q),$$

and relate the gauge freedom for $B = dA$ to the Aharonov–Bohm potential.

Exercise 47 (Liouville's theorem)
Prove that every Hamiltonian flow preserves the symplectic volume $\omega^n/n!$.

Exercise 48 (Phase volume produces ideal-gas thermodynamics)
For Liouville measure $d\mu_L = \omega^n/n!$, show that Hamiltonian evolution preserves the fine-grained Gibbs entropy

$$S_G[\rho] = -k_B \int_M \rho \log \rho \, d\mu_L.$$

For N indistinguishable noninteracting particles in a box, divide phase volume by $N!h^{3N}$ and use

$$H(q,p) = \sum_{a=1}^{N} \frac{|p_a|^2}{2m}.$$

Derive the Sackur–Tetrode entropy and the ideal-gas laws

$$E = \frac{3}{2} N k_B T, \qquad PV = N k_B T.$$

Derive the adiabatic and sound-speed laws, including, for $\varrho = mN/V$,

$$c = \sqrt{\frac{5P}{3\varrho}}.$$

Reconcile the entropy increase in free expansion with conservation of fine-grained Gibbs entropy.

For $f, g \in C^\infty(M)$, define

$$\{f, g\} = \omega(X_f, X_g).$$

Exercise 49 (Hamiltonian symmetry produces conservation laws)

Prove that a Hamiltonian f generates a symmetry of H exactly when f is conserved by the flow of H, and distinguish this statement from Noether's variational theorem [36].

Exercise 50 (Central forces and angular momentum)

On $T^*\mathbb{R}^3$, let

$$H(q,p) = \frac{|p|^2}{2m} + V(|q|), \qquad L = q \times p.$$

Derive the angular-momentum Poisson algebra and show that rotational symmetry confines each nonradial trajectory to a fixed plane.

2.8 Riemannian geometry

An **affine connection** is a connection on TM; write $\nabla_X Y = (\nabla Y)(X)$. For a covector field α, set

$$(\nabla_X \alpha)(Y) = X(\alpha(Y)) - \alpha(\nabla_X Y).$$

Define the covariant derivative of general tensor fields by the product rule and by requiring it to commute with contractions.

An affine connection is **torsion-free and metric-compatible** when

$$\nabla_X Y - \nabla_Y X = [X,Y], \qquad \nabla g = 0.$$

Such a connection is called a **Levi-Civita connection**.

Exercise 51 (Levi-Civita connection)

Prove that every pseudo-Riemannian manifold admits a unique Levi-Civita connection. Derive the coordinate-free Koszul formula.

An affinely parametrized curve $\gamma : I \to M$ is a **geodesic** if $\nabla_{\dot\gamma}\dot\gamma = 0$. The Lie derivative of the metric is

$$(\mathcal{L}_X g)(Y,Z) = X(g(Y,Z)) - g([X,Y],Z) - g(Y,[X,Z]),$$

and X is a **Killing vector field** if $\mathcal{L}_X g = 0$.

Let (M,g) be a pseudo-Riemannian manifold with Levi-Civita connection ∇.

Exercise 52 (Killing fields and conserved quantities)

Let X be a Killing vector field and let γ be an affinely parametrized geodesic. Prove that

$$\frac{d}{dt} g(\dot\gamma,\dot\gamma) = 0, \qquad \frac{d}{dt} g(X,\dot\gamma) = 0.$$

Deduce that $g(X,\dot\gamma)$ is constant along γ.

The **curvature** of ∇ is the $\mathrm{End}(TM)$-valued 2-form F_∇ determined by

$$R(X,Y)Z := F_\nabla(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z.$$

Exercise 53 (Curvature as a tensor)
For $f \in C^\infty(M)$ and vector fields X, Y, Z, prove that

$$R(fX,Y)Z = fR(X,Y)Z, \qquad R(X,fY)Z = fR(X,Y)Z,$$
$$R(X,Y)(fZ) = fR(X,Y)Z.$$

Prove that $R(X,Y) = -R(Y,X)$ and deduce that

$$R \in \Gamma\big(\Lambda^2 T^*M \otimes T^*M \otimes TM\big).$$

Exercise 54 (Geodesic deviation) *Let $\Gamma : (-\varepsilon,\varepsilon) \times I \to M$ be a smooth variation through affinely parametrized geodesics, and define the vector fields along Γ by*

$$T = \frac{\partial \Gamma}{\partial t}, \qquad J = \frac{\partial \Gamma}{\partial s}.$$

Prove that $[T,J] = 0$ and that

$$\nabla_T \nabla_T J = R(T,J)T.$$

Exercise 55 (Geodesic deviation produces interferometer strain)
Consider the weak plane-wave metric

$$g = -dt^2 + \big(1 + h_+(t-z)\big)dx^2 + \big(1 - h_+(t-z)\big)dy^2 + dz^2, \qquad |h_+| \ll 1.$$

Use geodesic deviation to show, to first order, that freely falling mirrors separated along the x- and y-axes acquire opposite fractional length changes $h_+/2$ and $-h_+/2$. Deduce that an equal-arm Michelson interferometer measures differential strain h_+, and compare this prediction with the peak strain of order 10^{-21} observed for GW150914 [38].

Exercise 56 (Gauss–Bonnet theorem)
Let (M,g) be a compact oriented Riemannian surface with piecewise smooth boundary. With Gaussian curvature K, signed geodesic curvature k_g, and signed exterior angles θ_j, prove [37]

$$\int_M K\,\mathrm{vol}_g + \int_{\partial M} k_g\,ds + \sum_j \theta_j = 2\pi\chi(M).$$

Derive its consequences for round spheres, positively curved tori, and closed hyperbolic surfaces.

The trace used below is the ordinary trace defined in the Algebra section. Define the **Ricci tensor**, **scalar curvature**, and **Einstein tensor** by

$$\mathrm{Ric}(X,Y) = \mathrm{tr}\big(Z \mapsto R(Z,X)Y\big), \qquad S = \mathrm{tr}_g(\mathrm{Ric}), \qquad G = \mathrm{Ric} - \frac{1}{2}Sg.$$

For a symmetric covariant two-tensor A, its **divergence** is the covector field

$$(\mathrm{div}\,A)(Y) = \mathrm{tr}_g\big((X,Z) \mapsto (\nabla_X A)(Z,Y)\big).$$

Exercise 57 (Contracted Bianchi identity and conservation)
Assume the second Bianchi identity

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0.$$

Contract this identity to prove that

$$\operatorname{div} G = 0.$$

Deduce that if T is a symmetric covariant two-tensor satisfying

$$G + \Lambda g = \kappa T$$

for constants Λ, κ with $\kappa \neq 0$, then $\operatorname{div} T = 0$.

2.9 Yang–Mills theory and the Standard Model

Let G be a Lie group with Lie algebra $\mathfrak{g}$. A **principal G-bundle** is a smooth free right G-manifold P with quotient $M = P/G$ such that the projection $P \to M$ is a locally trivial fiber bundle. For a finite-dimensional representation $\rho : G \to \mathrm{GL}(V)$, the **associated bundle** is

$$E = P \times_G V = (P \times V)/\sim, \qquad (p \cdot g, v) \sim (p, \rho(g)^{-1} v).$$

A **principal connection** is a $\mathfrak{g}$-valued one-form A on P that is G-equivariant and reproduces the generators of the fundamental vector fields. In a local trivialization it is represented by an ordinary form $A \in \Omega^1(U, \mathfrak{g})$, with curvature

$$F_A = dA + A \wedge A \in \Omega^2(U, \mathfrak{g}).$$

Exercise 58 (Isospin motivates Yang–Mills connections)
Treat the proton and neutron as a nucleon doublet carrying the defining representation of $\mathrm{SU}(2)$. *Gauge this symmetry and derive*

$$A \longmapsto A^g = g^{-1} A g + g^{-1} dg.$$

Show that the resulting connection differentiates associated-bundle fields and has covariant curvature, and relate this construction to [40].

Exercise 59 (Yang–Mills action)
On an oriented Riemannian four-manifold, consider the Yang–Mills action

$$S(A) = \int_M \mathrm{Tr}(F_A \wedge \star F_A).$$

Prove gauge invariance, derive

$$d_A \star F_A = 0,$$

and recover source-free Maxwell theory in the abelian case.

Exercise 60 (Standard Model symmetry breaking)
Take the Standard Model gauge group to be

$$G_{\mathrm{SM}} = \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y.$$

Write $(\mathbf{n},\mathbf{m})_Y$ *for the representation of dimension* n *under* $\mathrm{SU}(3)_C$*, dimension* m *under* $\mathrm{SU}(2)_L$*, and weak hypercharge* Y*. Take one generation of Weyl fermion multiplets to be*

$$Q_L = (\mathbf{3},\mathbf{2})_{1/6}, \qquad u_R = (\mathbf{3},\mathbf{1})_{2/3}, \qquad d_R = (\mathbf{3},\mathbf{1})_{-1/3},$$

$$L_L = (\mathbf{1},\mathbf{2})_{-1/2}, \qquad e_R = (\mathbf{1},\mathbf{1})_{-1},$$

and define $Q = T_3 + Y$*. Let* ϕ *be a section of the Higgs bundle associated to* $(\mathbf{1},\mathbf{2})_{1/2}$*, with*

$$V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4, \qquad D\phi = d\phi + \tfrac{i}{2}g\,W^a\sigma_a\phi + \tfrac{i}{2}g' B\phi,$$

where $\mu^2, \lambda > 0$ *and* g, g' *are coupling constants. From these data derive the fermion charges, the symmetry breaking* $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{U}(1)_{\mathrm{EM}}$*, the* W *and* Z *masses, the massless photon, and cancellation of the hypercharge anomalies. Identify the* $W^\pm \to e^\pm\nu$ *missing-transverse-energy and* $Z^0 \to \ell^+\ell^-$ *invariant-mass signatures permitted by these representations, and compare them with the original UA1 observations [41, 42].*

2.10 Chern–Simons theory and TQFT

Exercise 61 (Chern–Simons transgression)
For two connections on E*, set*

$$\nabla^t = (1-t)\nabla^0 + t\nabla^1, \qquad A = \nabla^1 - \nabla^0, \qquad F_t = F_{\nabla^t},$$

where $A \in \Omega^1(M, \operatorname{End} E)$*, and define*

$$\mathrm{CS}_P(\nabla^0,\nabla^1) = k\int_0^1 P(A, F_t, \ldots, F_t)\,dt.$$

Prove that

$$P(F_{\nabla^1}) - P(F_{\nabla^0}) = d\;\mathrm{CS}_P(\nabla^0,\nabla^1).$$

Let $\mathfrak{g} \subseteq M_N(\mathbb{C})$ be a matrix Lie algebra whose trace pairing is nondegenerate. For $A \in \Omega^1(M,\mathfrak{g})$, wedge products include matrix multiplication. As above, write

$$F_A = dA + A\wedge A$$

and define

$$\mathrm{CS}(A) = \mathrm{tr}\left(A\wedge dA + \frac{2}{3}A\wedge A\wedge A\right).$$

Exercise 62 (Chern–Simons form)
Prove that

$$d\ \mathrm{CS}(A) = \mathrm{tr}(F_A \wedge F_A).$$

For a closed oriented three-manifold M, define

$$S_{\mathrm{CS}}(A) = \int_M \mathrm{CS}(A).$$

Exercise 63 (Chern–Simons equation)
For $\alpha \in \Omega^1(M, \mathfrak{g})$, set $A_t = A + t\alpha$. Prove that

$$\left.\frac{d}{dt}\right|_{t=0} S_{\mathrm{CS}}(A_t) = 2\int_M \mathrm{tr}(\alpha \wedge F_A).$$

Deduce that the critical points satisfy $F_A = 0$.

For $k \in \mathbb{R}$, set $S_k = (k/4\pi)S_{\mathrm{CS}}$.

Exercise 64 (Gauge transformation)
Let G be a compact, connected, simple, and simply connected matrix Lie group with Lie algebra $\mathfrak{g}$. For a smooth map $g : M \to G$, define

$$A^g = g^{-1}Ag + g^{-1}dg.$$

Normalize the invariant trace so that

$$\nu(g) = \frac{1}{24\pi^2}\int_M \mathrm{tr}\big((g^{-1}dg)^3\big) \in \mathbb{Z}.$$

Prove that

$$\mathrm{CS}(A^g) - \mathrm{CS}(A) = -d\ \mathrm{tr}(dg\, g^{-1} \wedge A) - \frac{1}{3}\,\mathrm{tr}\big((g^{-1}dg)^3\big).$$

Deduce that $\exp(iS_k)$ is invariant under all gauge transformations when $k \in \mathbb{Z}$.

Now restrict to $2+1$ spacetime dimensions. Let M be a closed oriented three-manifold and let $\ell > 0$, so the cosmological constant is $\Lambda = -\ell^{-2}$. After identifying the coframe representation with $\mathfrak{sl}(2,\mathbb{R})$, let

$$e, \omega \in \Omega^1(M, \mathfrak{sl}(2,\mathbb{R})), \qquad A^{\pm} = \omega \pm \frac{1}{\ell}e.$$

The pair (A^+, A^-) has gauge Lie algebra $\mathfrak{sl}(2,\mathbb{R}) \oplus \mathfrak{sl}(2,\mathbb{R})$. Write

$$d_\omega e = de + \omega \wedge e + e \wedge \omega, \qquad F_\omega = d\omega + \omega \wedge \omega.$$

Exercise 65 (Flat connections encode three-dimensional gravity)
Prove that $F_{A^+} = F_{A^-} = 0$ is equivalent to

$$d_\omega e = 0, \qquad F_\omega + \frac{1}{\ell^2} e \wedge e = 0.$$

Assuming that $e : TM \to \mathfrak{sl}(2,\mathbb{R})$ is fiberwise invertible, identify these equations with the torsion-free condition and the Einstein equation with constant negative curvature.

Exercise 66 (Three-dimensional gravity is a Chern–Simons theory)
For the first-order Einstein–Hilbert functional

$$S_{\mathrm{EH}}(e,\omega) = \int_M \mathrm{tr}\left(e \wedge F_\omega + \frac{1}{3\ell^2} e \wedge e \wedge e\right).$$

Prove that, up to normalization and an exact term, it equals

$$S_{\mathrm{CS}}(A^+) - S_{\mathrm{CS}}(A^-)$$

[45, 46].

For a connection A and a finite-dimensional representation V of its gauge group, parallel transport around an oriented closed curve γ gives the holonomy $\mathrm{Hol}_\gamma(A)$. Define the **Wilson loop**

$$W_V(\gamma, A) = \mathrm{tr}_V(\mathrm{Hol}_\gamma(A)).$$

For a framed oriented link $L = \gamma_1 \sqcup \cdots \sqcup \gamma_r$ labeled by $V_1, \ldots, V_r$, set

$$W_L(A) = \prod_{j=1}^{r} W_{V_j}(\gamma_j, A).$$

For a framed link, define the formal Chern–Simons expectation value [43]

$$\langle W_L \rangle_k = \frac{\int e^{iS_k(A)} W_L(A)\, \mathcal{D}A}{\int e^{iS_k(A)}\, \mathcal{D}A}.$$

Exercise 67 (Wilson loops and the Jones polynomial)
For normalized $SU(2)$ theory at level k, set

$$q = \exp\left(\frac{2\pi i}{k+2}\right).$$

Assume the normalized Wilson-loop expectations satisfy

$$q^{-1}\langle W_{L_+}\rangle_k - q\langle W_{L_-}\rangle_k = \left(q^{1/2} - q^{-1/2}\right)\langle W_{L_0}\rangle_k$$

and $\langle W_{\bigcirc}\rangle_k = 1$. Show that they determine the Jones polynomial, account for framing dependence, and agree with the Reshetikhin–Turaev invariant of Exercise 29.

3 Analysis

3.1 Operators on Hilbert space

A **Hilbert space** is a complete complex inner-product space. Write $\mathcal{B}(H, K)$ for the bounded operators $H \to K$ with

$$\|T\| = \sup_{\|x\|=1} \|Tx\|, \qquad \mathcal{B}(H) = \mathcal{B}(H, H).$$

A C^***-algebra** is a complex Banach $*$-algebra A satisfying $\|a^*a\| = \|a\|^2$. An element $a \in A$ is **positive**, written $a \geq 0$, if $a = b^*b$ for some $b \in A$. The algebra $M_n(A)$ has its standard C^*-algebra structure.

Exercise 68 (Hilbert-space operators form a C^*-algebra)
Prove the Cauchy–Schwarz inequality, construct the adjoint of every $T \in \mathcal{B}(H,K)$, and prove

$$(ST)^* = T^*S^*, \qquad \|T^*T\| = \|T\|^2.$$

Deduce that $\mathcal{B}(H)$ is a C^*-algebra and that $M_n(\mathcal{B}(H)) \cong \mathcal{B}(H^{\oplus n})$ isometrically as a $*$-algebra.

3.2 Operator K-theory

For a unital C^*-algebra A, let $M_\infty(A) = \bigcup_{n\geq 1} M_n(A)$ under $x \mapsto x \oplus 0$. Projections $p, q \in M_\infty(A)$ are **stably Murray–von Neumann equivalent** if, after adjoining zero blocks, there is $v \in M_\infty(A)$ with

$$v^*v = p, \qquad vv^* = q.$$

Let $V(A)$ be the resulting set of equivalence classes with addition induced by direct sum. Define $K_0(A)$ as the Grothendieck group of $V(A)$.

Let $U_\infty(A) = \bigcup_{n\geq 1} U(M_n(A))$ under $u \mapsto u \oplus 1$, and define $K_1(A) = \pi_0(U_\infty(A))$. For a nonunital C^*-algebra A, let $\widetilde{A}$ be its unitization with quotient $\epsilon : \widetilde{A} \to \mathbb{C}$, and define

$$K_i(A) = \ker\Big(K_i(\widetilde{A}) \xrightarrow{\epsilon_*} K_i(\mathbb{C})\Big), \qquad i = 0, 1.$$

Exercise 69 (Stable projections and unitaries define K-groups)
Prove that direct sum makes $V(A)$ a commutative monoid, that $K_1(A)$ is an abelian group, and that K_0 and K_1 are functorial for $*$-homomorphisms. Prove the stable isomorphisms

$$K_i(M_n(A)) \cong K_i(A)$$

and compute

$$K_0(\mathbb{C}) \cong \mathbb{Z}, \qquad K_1(\mathbb{C}) = 0.$$

An operator $K \in \mathcal{B}(H,K')$ is **compact** if the image of its closed unit ball has compact closure. Write $\mathcal{K}(H)$ for the compact operators on H.

Exercise 70 (Compact operators carry integer-valued K-theory)
For an infinite-dimensional separable Hilbert space H, prove that $\mathcal{K}(H)$ is the norm-closed ideal generated by finite-rank operators. Classify projections in $M_\infty(\mathcal{K}(H))$ by rank and prove

$$K_0(\mathcal{K}(H)) \cong \mathbb{Z}, \qquad K_1(\mathcal{K}(H)) = 0.$$

For compact Hausdorff X, write $C(X)$ for the unital C^*-algebra of continuous complex-valued functions on X with pointwise operations.

Exercise 71 (Operator K-theory recovers vector bundles)
For compact Hausdorff X, associate to $p \in M_n(C(X))$ the bundle with fiber $\operatorname{im} p(x)$. Prove the Serre–Swan equivalence between complex vector bundles over X and finitely generated projective $C(X)$-modules, where projective means isomorphic to a direct summand of $C(X)^n$ [8]. Deduce the natural isomorphism

$$K_0(C(X)) \cong K^0(X).$$

Identify this description with the projection-valued classification of Exercise 38.

3.3 Fredholm operators and index theory

An operator $T : H \to K'$ is **Fredholm** if its kernel and cokernel are finite dimensional and its image is closed; define

$$\operatorname{ind}(T) = \dim \ker T - \dim \operatorname{coker} T.$$

Exercise 72 (The Calkin boundary map is the Fredholm index)
Let H be infinite-dimensional and separable and set $\mathcal{Q}(H) = \mathcal{B}(H)/\mathcal{K}(H)$. Prove that $T \in \mathcal{B}(H)$ is Fredholm exactly when its image $q(T) \in \mathcal{Q}(H)$ is invertible. Construct the six-term K-theory sequence of [7]

$$0 \longrightarrow \mathcal{K}(H) \longrightarrow \mathcal{B}(H) \xrightarrow{q} \mathcal{Q}(H) \longrightarrow 0$$

Compute $K_i(\mathcal{B}(H)) = 0$, use polar retraction to regard $q(T)$ as a K_1-class, and prove, with the index convention for the boundary map,

$$\partial : K_1(\mathcal{Q}(H)) \xrightarrow{\cong} K_0(\mathcal{K}(H)) \cong \mathbb{Z}, \qquad \partial[q(T)] = [P_{\ker T}] - [P_{\ker T^*}] = \operatorname{ind}(T).$$

Deduce that the Fredholm index is locally constant, additive under composition, and invariant under compact perturbations.

Let $\operatorname{Fred}(H)$ denote the Fredholm operators on H, with the operator-norm topology.

Exercise 73 (Fredholm families represent K-theory)
For an infinite-dimensional separable Hilbert space H and compact Hausdorff space X, use Exercise 72 to construct the index bundle of a norm-continuous family $X \to \operatorname{Fred}(H)$ and prove the Atiyah–Jänich isomorphism [1]

$$[X, \operatorname{Fred}(H)] \cong K^0(X).$$

Let M be a closed smooth manifold and let $E, F \to M$ be complex vector bundles. An order-m **differential operator** $D : \Gamma(E) \to \Gamma(F)$ is locally of the form $D = \sum_{|\alpha| \le m} a_\alpha(x)\partial^\alpha$. Its **principal symbol** is

$$\sigma_D(x,\xi) = \sum_{|\alpha|=m} a_\alpha(x)\xi^\alpha : E_x \longrightarrow F_x.$$

The operator is **elliptic** if $\sigma_D(x,\xi)$ is invertible for $\xi \neq 0$. Write $K_c^0(Y) = \widetilde{K}^0(Y^+)$ for compactly supported K-theory. The **symbol class** of an elliptic operator is

$$[\sigma_D] = [\pi^*E, \pi^*F, \sigma_D] \in K_c^0(T^*M).$$

Define $\mathrm{ind}_{\mathrm{top}} : K_c^0(T^*M) \to \mathbb{Z}$ by the K-theoretic pushforward to a point.

Exercise 74 (Atiyah–Singer index theorem)

Prove that an elliptic operator D has a Fredholm extension and that [2]

$$\mathrm{ind}(D) = \dim\ker D - \dim\ker D^* = \mathrm{ind}_{\mathrm{top}}[\sigma_D].$$

For a Riemannian metric on M, apply this to the de Rham operator $d + d^* : \Omega^{\mathrm{even}}(M) \to \Omega^{\mathrm{odd}}(M)$ and recover $\mathrm{ind}(d+d^*) = \chi(M)$.

Exercise 75 (Instanton charge produces the chiral anomaly)

Let A be a finite-action anti-self-dual $\mathrm{SU}(N)$ connection on $\mathbb{R}^4$ extending to a Hermitian bundle $E \to S^4$. For skew-Hermitian curvature F_A and the fundamental trace, define its topological charge by

$$Q(A) = \frac{1}{8\pi^2}\int_{S^4} \mathrm{tr}(F_A \wedge F_A) = \langle c_2(E), [S^4]\rangle.$$

Prove that $Q(A) \in \mathbb{Z}$. Let $S^{\pm}$ be the half-spinor bundles and $D_A^{\pm}$ the coupled chiral Dirac operators. Use Exercise 74 to prove

$$\dim\ker D_A^- - \dim\ker D_A^+ = Q(A).$$

For N_f massless Dirac fermions in the fundamental representation, deduce the integrated chiral anomaly [3]

$$\Delta Q_5 = \int_{\mathbb{R}^4} d(\star J_5) = 2N_f Q(A).$$

3.4 Operator spaces and completely bounded maps

A **concrete operator space** is a closed linear subspace $E \subseteq \mathcal{B}(H,K)$. Its nth matrix level is

$$M_n(E) \subseteq \mathcal{B}(H^{\oplus n}, K^{\oplus n})$$

with the inherited operator norm. For a linear map $\phi : E \to F$, define

$$\phi_n([x_{ij}]) = [\phi(x_{ij})], \qquad \|\phi\|_{\mathrm{cb}} = \sup_{n\geq 1}\|\phi_n\|.$$

The map is a **complete contraction**, **complete isometry**, or **completely bounded** when every ϕ_n is contractive, isometric, or uniformly bounded, respectively. Write $\mathrm{CB}(E,F)$ for the completely bounded maps.

Exercise 76 (Ruan's axioms characterize concrete operator spaces)
Show that concrete matrix norms satisfy

$$\|x \oplus y\| = \max\{\|x\|, \|y\|\}, \qquad \|\alpha x \beta\| \leq \|\alpha\| \, \|x\| \, \|\beta\|.$$

Conversely, prove that a vector space with matrix norms satisfying these axioms admits a completely isometric embedding into $\mathcal{B}(H)$ [9].

The **row** and **column operator spaces** are

$$R_n = M_{1,n} \subseteq \mathcal{B}(\mathbb{C}^n, \mathbb{C}), \qquad C_n = M_{n,1} \subseteq \mathcal{B}(\mathbb{C}, \mathbb{C}^n).$$

Exercise 77 (Matrix levels detect noncommutativity)
For the identity maps between the underlying Hilbert spaces of R_n and C_n, prove

$$\|1 : R_n \to C_n\|_{\mathrm{cb}} = \|1 : C_n \to R_n\|_{\mathrm{cb}} = \sqrt{n}.$$

For transposition $t : M_n \to M_n$, prove

$$\|t\| = 1, \qquad \|t\|_{\mathrm{cb}} = n.$$

Explain why Banach-space norm alone forgets both distinctions.

Exercise 78 (Smith's lemma stabilizes maps into matrices)
For an operator space E and a linear map $\phi : E \to M_n$, prove [10]

$$\|\phi\|_{\mathrm{cb}} = \|\phi_n\|.$$

Deduce that complete boundedness is a finite computation whenever the codomain is a fixed matrix algebra.

3.5 Operator-space duality and Haagerup tensoring

The **operator-space dual** of E is the Banach dual E^* with matrix norms determined by

$$M_n(E^*) = \mathrm{CB}(E, M_n)$$

isometrically. Rectangular spaces $M_{p,q}(E)$ carry the norms inherited from $\mathcal{B}(H^{\oplus q}, K^{\oplus p})$. For $u = \sum_{k=1}^{r} x_k \otimes y_k \in E \otimes F$, define

$$\|u\|_h = \inf\left\{ \|[x_1 \ \cdots \ x_r]\|_{M_{1,r}(E)} \|[y_1 \ \cdots \ y_r]^{\mathsf{T}}\|_{M_{r,1}(F)} \right\}.$$

The completion is the **Haagerup tensor product** $E \otimes_h F$.

Exercise 79 (Rows and columns give Schatten norms)
Under $e_i \otimes e_j \mapsto E_{ij}$, prove the complete isometries

$$C_n \otimes_h R_n \cong M_n, \qquad R_n \otimes_h C_n \cong S_1^n,$$

where M_n has operator norm and S_1^n has trace norm. Interpret the reversal of the tensor factors as the distinction between largest singular value and total singular mass.

Exercise 80 (Trace norm governs state discrimination)
Let $\rho_0, \rho_1 \in M_n$ be positive with trace one, and let p_0, p_1 be probabilities with $p_0 + p_1 = 1$. A binary measurement is a pair $M_0, M_1 \geq 0$ with $M_0 + M_1 = 1$. Define $\|x\|_1 = \operatorname{tr}\sqrt{x^*x}$. For

$$p_{\text{succ}} = p_0 \operatorname{tr}(M_0\rho_0) + p_1 \operatorname{tr}(M_1\rho_1),$$

prove the Helstrom formula [12]

$$\max_{(M_0,M_1)} p_{\text{succ}} = \frac{1}{2}\left(1 + \|p_0\rho_0 - p_1\rho_1\|_1\right).$$

Determine an optimal measurement from the positive spectral subspace of $p_0\rho_0 - p_1\rho_1$, and use $R_n \otimes_h C_n \cong S_1^n$ to interpret distinguishability as a Haagerup tensor norm.

For a bilinear map $u : E \times F \to G$, define

$$u_n(x,y)_{ij} = \sum_{k=1}^{n} u(x_{ik}, y_{kj}), \qquad \|u\|_{\text{mb}} = \sup_n \|u_n\|.$$

Such a map is **multiplicatively bounded** if this supremum is finite.

Exercise 81 (Haagerup tensoring linearizes products)
Prove that u is multiplicatively bounded exactly when its linearization extends to a completely bounded map

$$\widetilde{u} : E \otimes_h F \longrightarrow G, \qquad \widetilde{u}(x \otimes y) = u(x,y),$$

and prove $\|\widetilde{u}\|_{\text{cb}} = \|u\|_{\text{mb}}$. Apply this to multiplication in a C^*-algebra.

3.6 Operator systems and completely positive maps

An **operator system** is a self-adjoint unital subspace $S \subseteq \mathcal{B}(H)$. A linear map $\Phi : S \to T$ is **positive** if $\Phi(S_+) \subseteq T_+$, and **completely positive** if every $\Phi_n : M_n(S) \to M_n(T)$ is positive.

Exercise 82 (Partial transpose detects entanglement)
A state in $M_m \otimes M_n$ is a positive element ρ with $\operatorname{tr}\rho = 1$. Call it separable if $\rho = \sum_k p_k \sigma_k \otimes \tau_k$ for states σ_k, τ_k and probabilities p_k. Prove that transposition $t : M_n \to M_n$ is positive and that every separable state has positive partial transpose [13]

$$\rho^\Gamma = (1 \otimes t)(\rho) \geq 0.$$

For $\Omega_2 = (|00\rangle + |11\rangle)/\sqrt{2}$, compute the spectrum of $(|\Omega_2\rangle\langle\Omega_2|)^\Gamma$ and deduce that t is not 2-positive. Assume the Peres–Horodecki theorem that positivity of the partial transpose is also sufficient for separability in 2×2 and 2×3 systems [14]. Apply it to

$$\rho_p = p|\psi^-\rangle\langle\psi^-| + (1-p)\frac{1_4}{4}, \qquad |\psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}},$$

and prove that ρ_p is separable exactly when $p \leq 1/3$.

For $\Phi : M_n \to M_m$, its **Choi matrix** is

$$C_\Phi = \sum_{i,j=1}^{n} E_{ij} \otimes \Phi(E_{ij}) \in M_n \otimes M_m.$$

Exercise 83 (Choi matrices classify completely positive maps)
Prove that the following are equivalent [11]:

$$\Phi \text{ is completely positive}, \qquad C_\Phi \geq 0, \qquad \Phi(x) = \sum_\alpha V_\alpha x V_\alpha^*.$$

Determine the least number of Kraus operators from $\operatorname{rank} C_\Phi$ *and express trace preservation and unitality as partial-trace conditions on* C_Φ*.*

Exercise 84 (Choi states enable process tomography)
Let $\Omega_n = n^{-1/2} \sum_{i=1}^{n} e_i \otimes e_i$ *and let* $\Phi : M_n \to M_m$ *be completely positive and trace preserving. Define*

$$\rho_\Phi = (1 \otimes \Phi)(|\Omega_n\rangle\langle\Omega_n|).$$

Prove

$$\rho_\Phi = \frac{1}{n} C_\Phi, \qquad \Phi(x) = n \ \mathrm{Tr}_1\big((x^\mathsf{T} \otimes 1)\rho_\Phi\big), \qquad \mathrm{Tr}_2\, \rho_\Phi = \frac{1}{n} 1.$$

Here Tr_1 *and* Tr_2 *are the partial traces over the first and second tensor factors. Call a bipartite probe* $\sigma \in M_n \otimes M_n$ *full operator-Schmidt rank if* $\sigma = \sum_{k=1}^{n^2} A_k \otimes B_k$ *with both families linearly independent. Prove that* $(1 \otimes \Phi)(\sigma)$ *determines every linear map* $\Phi : M_n \to M_m$ *exactly when* σ *has full operator-Schmidt rank. Construct a separable probe with this property, and compare the entangled- and separable-probe process reconstructions reported in the photonic experiment of Altepeter et al. [15].*

Exercise 85 (Stinespring dilation turns positivity into geometry)
For a completely positive map $\Phi : A \to \mathcal{B}(H)$ *from a unital* C^**-algebra, construct a* $*$*-representation* $\pi : A \to \mathcal{B}(K)$ *and* $V : H \to K$ *such that [32]*

$$\Phi(a) = V^* \pi(a) V, \qquad V^*V = \Phi(1).$$

Prove uniqueness of a minimal dilation up to unitary equivalence.

For $E \subseteq \mathcal{B}(H, K)$, define its **Paulsen operator system**

$$\mathcal{S}(E) = \left\{ \begin{pmatrix} \lambda 1_K & x \\ y^* & \mu 1_H \end{pmatrix} : \lambda, \mu \in \mathbb{C},\ x, y \in E \right\} \subseteq \mathcal{B}(K \oplus H).$$

Exercise 86 (The 2×2 trick converts contractions into positivity)
For $\phi : E \to \mathcal{B}(H_1, H_2)$*, prove that* ϕ *is a complete contraction exactly when the unital map*

$$\begin{pmatrix} \lambda 1_K & x \\ y^* & \mu 1_H \end{pmatrix} \longmapsto \begin{pmatrix} \lambda 1_{H_2} & \phi(x) \\ \phi(y)^* & \mu 1_{H_1} \end{pmatrix}$$

is completely positive. Use this operator-system encoding to derive the completely contractive extension theorem.

Exercise 87 (Wittstock factorizes completely bounded maps)
For a completely bounded map $\phi : A \to \mathcal{B}(H)$, prove that there are a representation $\pi : A \to \mathcal{B}(K)$ and operators $V, W : H \to K$ such that [16]

$$\phi(a) = V^*\pi(a)W, \qquad \|\phi\|_{\mathrm{cb}} = \inf\{\|V\|\|W\| : \phi(\cdot) = V^*\pi(\cdot)W\}.$$

Deduce that every completely bounded map is a linear combination of completely positive maps.

3.7 Quantum channels and matrix-level distinguishability

For $x \in M_n$, write $\|x\|_1 = \operatorname{tr}\sqrt{x^*x}$. A **state** is $\rho \in M_n$ with $\rho \geq 0$ and $\operatorname{tr}\rho = 1$. The **partial trace** $\mathrm{Tr}_B : M_n \otimes M_m \to M_n$ is characterized by

$$\operatorname{tr}(a\ \mathrm{Tr}_B\, x) = \operatorname{tr}((a \otimes 1)x)\,.$$

A **Schrödinger-picture quantum channel** is a completely positive trace-preserving map $\Phi : M_n \to M_m$. Its **Heisenberg adjoint** $\Phi^* : M_m \to M_n$ is determined by

$$\operatorname{tr}(\Phi(\rho)a) = \operatorname{tr}(\rho\Phi^*(a)).$$

Exercise 88 (Schmidt spectra measure entanglement)
For $\psi \in H_A \otimes H_B$, apply singular-value decomposition to $T_\psi : H_A^* \to H_B$, where $T_\psi(f) = (f \otimes 1)(\psi)$, and obtain

$$\psi = \sum_{i=1}^{r} s_i a_i \otimes b_i.$$

Characterize product vectors by $r = 1$, identify s_i^2 as the nonzero eigenvalues of both reduced states, and compute

$$S(\psi) = -\sum_i s_i^2 \log s_i^2.$$

Exercise 89 (Channels are isometries with an environment)
Combine Choi's theorem and Stinespring dilation to prove that every channel has equivalent forms

$$\Phi(\rho) = \sum_\alpha K_\alpha \rho K_\alpha^* = \mathrm{Tr}_E(V\rho V^*), \qquad V^*V = 1.$$

Describe the Choi–Jamiołkowski correspondence and determine which bipartite states correspond to channels.

For a linear map $\Psi : M_n \to M_m$, define its **diamond norm** by

$$\|\Psi\|_\diamond = \sup_{k\geq 1} \|\Psi \otimes 1_{M_k}\|_{1\to 1}.$$

Exercise 90 (The diamond norm is the operational cb norm)
Prove the stabilization and duality identities [17]

$$\|\Psi\|_\diamond = \|\Psi \otimes 1_{M_n}\|_{1\to 1}, \qquad \|\Psi\|_\diamond = \|\Psi^*\|_{\mathrm{cb}}.$$

For equally likely channels Φ_0, Φ_1, prove that the optimal one-use discrimination probability, allowing an entangled ancilla, is

$$p_{\mathrm{succ}} = \frac{1}{2} + \frac{1}{4}\|\Phi_0 - \Phi_1\|_\diamond.$$

Exercise 91 (Entanglement improves process discrimination)
Assume the Piani–Watrous theorem that every entangled bipartite state improves the success probability in some channel-discrimination task [18]. Explain why this is an operational manifestation of matrix amplification. Reconstruct the subchannels and unsteerable benchmark in the photonic experiment of Sun et al., and compare its measured steering-assisted success probabilities with that benchmark [19].

Exercise 92 (No channel universally clones pure states)
Suppose a channel C and fixed blank state σ satisfy $C(\rho \otimes \sigma) = \rho \otimes \rho$ for every pure state ρ. Use contractivity of trace distance, or monotonicity of fidelity, to obtain a contradiction from two nonorthogonal states. Recover the unitary inner-product argument of Wootters and Zurek [31].

3.8 Nonlocal correlations and Grothendieck inequalities

Exercise 93 (Every entangled pure two-qubit state violates CHSH)
Let A_0, A_1 and B_0, B_1 be self-adjoint operators with eigenvalues in $\{-1, 1\}$, and define the CHSH operator

$$\mathcal{C} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1).$$

Establish the classical CHSH bound 2. For the two-qubit state

$$\psi_\theta = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle, \qquad 0 \le \theta \le \frac{\pi}{4},$$

derive the maximal CHSH value [27]

$$2\sqrt{1 + \sin^2(2\theta)}.$$

Conclude that every entangled pure two-qubit state violates CHSH. Relate the four observables to spacelike-separated measurement settings and compare the quantum violation with a loophole-free Bell experiment using electron spins separated by 1.3 kilometers [28].

A **bipartite binary quantum correlation matrix** has entries

$$c_{ij} = \langle \psi, (A_i \otimes B_j)\psi \rangle,$$

where ψ is a unit vector. The operators A_i and B_j are self-adjoint unitaries on finite-dimensional Hilbert spaces.

For a real matrix $M = (M_{ij})$, define

$$\beta_{\mathrm{C}}(M) = \max_{\varepsilon_i, \delta_j \in \{-1,1\}} \left| \sum_{i,j} M_{ij} \varepsilon_i \delta_j \right|$$

and

$$\beta_{\mathrm{Q}}(M) = \sup_c \left| \sum_{i,j} M_{ij} c_{ij} \right|,$$

where the supremum is over bipartite binary quantum correlation matrices.

Exercise 94 (Tsirelson turns quantum correlations into vectors)

Prove that every binary quantum correlation has a representation [29]

$$c_{ij} = \langle x_i, y_j \rangle_{\mathbb{R}}, \qquad \|x_i\|, \|y_j\| \leq 1.$$

Prove the converse using finite-dimensional Clifford representations and deduce the sharp CHSH value $2\sqrt{2}$.

The **real Grothendieck constant** $K_G^{\mathbb{R}}$ is the least constant for which

$$\beta_{\mathrm{Q}}(M) \leq K_G^{\mathbb{R}} \beta_{\mathrm{C}}(M)$$

holds for every real matrix M.

Exercise 95 (Grothendieck bounds quantum advantage uniformly)

Prove that $K_G^{\mathbb{R}} < \infty$ [30]. Use

$$M_{\mathrm{CHSH}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

to prove $K_G^{\mathbb{R}} \geq \sqrt{2}$ and interpret the inequality as a dimension-independent comparison of scalar and Hilbert-space strategies.

For a bilinear form $u : A \times B \to \mathbb{C}$, define $T_u : A \to B^*$ by $T_u(a)(b) = u(a, b)$. The form is **jointly completely bounded** when T_u is completely bounded, with $\|u\|_{\mathrm{jcb}} = \|T_u\|_{\mathrm{cb}}$. A **state** on a unital C^*-algebra is a positive functional of norm one.

Exercise 96 (The operator-space Grothendieck inequality)

Prove that for every jointly completely bounded $u : A \times B \to \mathbb{C}$ there are states f_1, f_2 on A and g_1, g_2 on B such that [20]

$$|u(a, b)| \leq \|u\|_{\mathrm{jcb}} \left(f_1(aa^*)^{1/2} g_1(b^*b)^{1/2} + f_2(a^*a)^{1/2} g_2(bb^*)^{1/2} \right).$$

Deduce a factorization of T_u through a direct sum of row and column Hilbert operator spaces, and recover the scalar Grothendieck pattern on commutative subalgebras.

3.9 Dual operator spaces and von Neumann algebras

For $\mathcal{S} \subseteq \mathcal{B}(H)$, its **commutant** is

$$\mathcal{S}' = \{T \in \mathcal{B}(H) : Tx = xT \text{ for every } x \in \mathcal{S}\}.$$

A **von Neumann algebra** is a unital $*$-subalgebra $M \subseteq \mathcal{B}(H)$ satisfying $M = M''$. A net (T_λ) converges to T in the **strong operator topology** if $T_\lambda \xi \to T\xi$ for every $\xi \in H$, and in the **weak operator topology** if

$$\langle T_\lambda \xi, \eta\rangle \longrightarrow \langle T\xi, \eta\rangle$$

for every $\xi, \eta \in H$.

Exercise 97 (Bicommutants are dual operator algebras)
Let $A \subseteq \mathcal{B}(H)$ be a unital $*$-subalgebra. Prove that

$$A'' = \overline{A}^{\mathrm{SOT}} = \overline{A}^{\mathrm{WOT}}.$$

Identify the canonical predual of A'' as a quotient of the trace-class operators.

An operator space E is a **dual operator space** with predual E_* if $E = E_*^*$ completely isometrically. A map between dual operator spaces is **normal** if it is weak-$*$ continuous. The **center** of a von Neumann algebra is $Z(M) = M \cap M'$, and M is a **factor** if $Z(M) = \mathbb{C}1$. A **normal channel** is a normal unital completely positive map.

Exercise 98 (Normal CP maps have Stinespring dilations)
Let M be a von Neumann algebra and let $\Phi : M \to \mathcal{B}(H)$ be normal and completely positive. Prove Stinespring's theorem [32]: there are a Hilbert space K, a normal representation $\pi : M \to \mathcal{B}(K)$, and $V : H \to K$ such that

$$\Phi(x) = V^*\pi(x)V, \qquad V^*V = \Phi(1).$$

Deduce that every normal channel on $\mathcal{B}(H_0)$ has the form

$$\Phi(x) = V^*(x \otimes 1_E)V$$

and that its preadjoint is a Schrödinger-picture channel.

A positive functional ψ is **normal** if it preserves suprema of bounded increasing nets of positive elements and **faithful** if $\psi(a^*a) = 0$ implies $a = 0$. A **type** II_1 **factor** is an infinite-dimensional factor with a faithful normal tracial state

$$\tau : M \longrightarrow \mathbb{C}, \qquad \tau(ab) = \tau(ba), \qquad \tau(1) = 1.$$

For projections $p, q \in M$, write $p \sim q$ if there is $v \in M$ such that

$$v^*v = p, \qquad vv^* = q.$$

Write $p \precsim q$ if p is equivalent to a projection r satisfying $rq = r$.

Exercise 99 (Projection dimension)

Let M be a type II_1 factor with normalized trace τ. Prove that

$$p \sim q \quad \Longrightarrow \quad \tau(p) = \tau(q),$$

and prove the comparison theorem

$$p \precsim q \quad \Longleftrightarrow \quad \tau(p) \leq \tau(q).$$

Deduce that $\tau(p) = \tau(q)$ if and only if $p \sim q$.

3.10 Conditional expectations and subfactors

A von Neumann algebra is **finite** if $v^*v = 1$ implies $vv^* = 1$. Let $N \subseteq M$ be finite von Neumann algebras with a common faithful normal tracial state τ. A **trace-preserving conditional expectation** is a normal unital completely positive map $E_N : M \to N$ such that

$$\tau \circ E_N = \tau, \qquad E_N(n_1 x n_2) = n_1 E_N(x) n_2$$

for $x \in M$ and $n_1, n_2 \in N$. Let $L^2(M, \tau)$ be the completion of M for

$$\langle x, y \rangle_\tau = \tau(y^* x).$$

Let M act on $L^2(M, \tau)$ by left multiplication.

Exercise 100 (Norm-one projections are conditional expectations)

Let A be a unital C^*-algebra, $B \subseteq A$ a unital C^*-subalgebra, and $P : A \to B$ a linear projection with $\|P\| = 1$. Prove Tomiyama's theorem [21]: P is positive, completely positive, and

$$P(b_1 a b_2) = b_1 P(a) b_2.$$

Exercise 101 (Conditional expectations give Jones projections)

Prove that there is a unique trace-preserving conditional expectation $E_N : M \to N$ and that it extends to the orthogonal projection

$$e_N : L^2(M, \tau) \longrightarrow L^2(N, \tau).$$

Prove that

$$e_N = e_N^* = e_N^2, \qquad e_N x e_N = E_N(x) e_N$$

for every $x \in M$.

The **basic construction** is

$$M_1 = \langle M, e_N \rangle'' \subseteq \mathcal{B}(L^2(M, \tau)),$$

the double commutant of the algebra generated by M and e_N. For a projection $p = [p_{ij}] \in M_k(N)$ acting by left multiplication, define

$$\dim_N\big(pL^2(N, \tau)^k\big) = \sum_{i=1}^{k} \tau(p_{ii}).$$

An inclusion $N \subseteq M$ has **finite Jones index** if $L^2(M,\tau)$ is isomorphic as a right N-module to such a module; then

$$[M:N] = \dim_N L^2(M,\tau).$$

Exercise 102 (The Pimsner–Popa bound detects finite index)
For an inclusion $N \subseteq M$ of type II_1 factors with trace-preserving expectation E_N, prove that finite index is equivalent to the existence of $\lambda > 0$ satisfying [22]

$$E_N(x) \geq \lambda x \qquad (x \in M_+).$$

Prove that the largest such λ is $[M:N]^{-1}$ and interpret this order inequality as a quantitative strengthening of complete positivity.

Exercise 103 (Jones basic construction)
Let $N \subseteq M$ be an inclusion of type II_1 factors of finite Jones index. Prove that M_1 is a finite factor and that its normalized trace satisfies [23]

$$\tau_1(xe_N y) = \frac{1}{[M:N]}\tau(xy)$$

for $x, y \in M$.

The **Jones tower** is the sequence obtained by iterating the basic construction:

$$N = M_{-1} \subset M = M_0 \subset M_1 \subset M_2 \subset \cdots .$$

Let e_i be the Jones projection used to construct M_i from M_{i-1}, and set

$$\delta = \sqrt{[M:N]}.$$

The **Temperley–Lieb algebra** with loop parameter δ is generated by U_i subject to

$$U_i^2 = \delta U_i, \qquad U_i U_j = U_j U_i \quad (|i-j| \geq 2), \qquad U_i U_{i\pm 1} U_i = U_i.$$

Exercise 104 (Temperley–Lieb relations)
Prove that the Jones projections satisfy [26, 24]

$$e_i^2 = e_i = e_i^*, \qquad e_i e_j = e_j e_i \quad (|i-j| \geq 2),$$

$$e_i e_{i\pm 1} e_i = \delta^{-2} e_i.$$

Deduce that $U_i \mapsto \delta e_i$ defines a representation of the Temperley–Lieb algebra.

Exercise 105 (Braid representation)
Let $q \in \mathbb{C}^\times$ satisfy $\delta = q + q^{-1}$ and define $\sigma_i = q\,1 - U_i$. Prove that σ_i is invertible with inverse $q^{-1}1 - U_i$, and prove that

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1},$$

$$\sigma_i \sigma_j = \sigma_j \sigma_i \qquad (|i-j| \geq 2).$$

Deduce that the σ_i define a representation of the braid group [25].

A **Markov trace** on the Temperley–Lieb tower is a compatible family of positive normalized traces satisfying

$$\operatorname{tr}(xU_i) = \delta^{-1}\operatorname{tr}(x)$$

whenever x belongs to the algebra generated by the preceding generators.

Exercise 106 (Positivity quantizes subfactor indices below four)
Define

$$P_0(\delta) = 1, \qquad P_1(\delta) = \delta, \qquad P_{n+1}(\delta) = \delta P_n(\delta) - P_{n-1}(\delta).$$

Prove that the normalized traces on the Jones tower restrict to a Markov trace. Prove that, when $\sin\theta \neq 0$*,*

$$P_n(2\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}.$$

Use positivity of the Markov trace to prove that if $0 < \delta < 2$*, then*

$$\delta = 2\cos\left(\frac{\pi}{m}\right)$$

for some integer $m \geq 3$*.*

Exercise 107 (Jones index theorem)
Let $N \subseteq M$ *be an inclusion of type* II_1 *factors of finite Jones index. Prove that [24]*

$$[M:N] \in \left\{4\cos^2\left(\frac{\pi}{n}\right) : n \geq 3\right\} \cup [4,\infty).$$

Relate the discrete values to the $SU(2)_k$ *braid representations and Chern–Simons Wilson-loop invariants constructed above.*

References

[1] Michael F. Atiyah, *K-Theory*, W. A. Benjamin, 1967.

[2] Michael F. Atiyah and Isadore M. Singer, *The index of elliptic operators I*, Annals of Mathematics, 87(3):484–530, 1968. doi:10.2307/1970715.

[3] Kazuo Fujikawa, *Path-integral measure for gauge-invariant fermion theories*, Physical Review Letters, 42(18):1195–1198, 1979. doi:10.1103/PhysRevLett.42.1195.

[4] Alexei Kitaev, *Periodic table for topological insulators and superconductors*, AIP Conference Proceedings, 1134:22–30, 2009. doi:10.1063/1.3149495.

[5] K. von Klitzing, G. Dorda, and M. Pepper, *New Method for High-Accuracy Determination of the Fine-Structure Constant Based on Quantized Hall Resistance*, Physical Review Letters, 45(6):494–497, 1980. doi:10.1103/PhysRevLett.45.494.

[6] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs, *Quantized Hall Conductance in a Two-Dimensional Periodic Potential*, Physical Review Letters, 49(6):405–408, 1982. doi:10.1103/PhysRevLett.49.405.

[7] Bruce Blackadar, *K-Theory for Operator Algebras*, 2nd ed., Mathematical Sciences Research Institute Publications, Vol. 5, Cambridge University Press, 1998.

[8] Richard G. Swan, *Vector bundles and projective modules*, Transactions of the American Mathematical Society, 105(2):264–277, 1962. doi:10.1090/S0002-9947-1962-0143225-6.

[9] Zhong-Jin Ruan, *Subspaces of C^*-algebras*, Journal of Functional Analysis, 76(1):217–230, 1988. doi:10.1016/0022-1236(88)90057-2.

[10] R. R. Smith, *Completely bounded maps between C^*-algebras*, Journal of the London Mathematical Society, s2-27(1):157–166, 1983. doi:10.1112/jlms/s2-27.1.157.

[11] Man-Duen Choi, *Completely positive linear maps on complex matrices*, Linear Algebra and its Applications, 10(3):285–290, 1975. doi:10.1016/0024-3795(75)90075-0.

[12] Carl W. Helstrom, *Quantum Detection and Estimation Theory*, Mathematics in Science and Engineering, Vol. 123, Academic Press, 1976.

[13] Asher Peres, *Separability criterion for density matrices*, Physical Review Letters, 77(8):1413–1415, 1996. doi:10.1103/PhysRevLett.77.1413.

[14] Michał Horodecki, Paweł Horodecki, and Ryszard Horodecki, *Separability of mixed states: necessary and sufficient conditions*, Physics Letters A, 223(1–2):1–8, 1996. doi:10.1016/S0375-9601(96)00706-2.

[15] J. B. Altepeter, D. Branning, E. Jeffrey, T. C. Wei, P. G. Kwiat, R. T. Thew, J. L. O'Brien, M. A. Nielsen, and A. G. White, *Ancilla-assisted quantum process tomography*, Physical Review Letters, 90(19):193601, 2003. doi:10.1103/PhysRevLett.90.193601.

[16] Gerd Wittstock, *Ein operatorwertiger Hahn–Banach Satz*, Journal of Functional Analysis, 40(2):127–150, 1981. doi:10.1016/0022-1236(81)90064-1.

[17] John Watrous, *Semidefinite programs for completely bounded norms*, Theory of Computing, 5(11):217–238, 2009. doi:10.4086/toc.2009.v005a011.

[18] Marco Piani and John Watrous, *All entangled states are useful for channel discrimination*, Physical Review Letters, 102(25):250501, 2009. doi:10.1103/PhysRevLett.102.250501.

[19] Kai Sun, Xiang-Jun Ye, Ya Xiao, Xiao-Ye Xu, Yu-Chun Wu, Jin-Shi Xu, Jing-Ling Chen, Chuan-Feng Li, and Guang-Can Guo, *Demonstration of Einstein–Podolsky–Rosen steering with enhanced subchannel discrimination*, npj Quantum Information, 4:12, 2018. doi:10.1038/s41534-018-0067-1.

[20] Uffe Haagerup and Magdalena Musat, *The Effros–Ruan conjecture for bilinear forms on C^*-algebras*, Inventiones Mathematicae, 174(1):139–163, 2008. doi:10.1007/s00222-008-0137-7.

[21] Jun Tomiyama, *On the projection of norm one in W^*-algebras*, Proceedings of the Japan Academy, 33(10):608–612, 1957. doi:10.3792/pja/1195524885.

[22] Mihai Pimsner and Sorin Popa, *Entropy and index for subfactors*, Annales scientifiques de l'École Normale Supérieure, 19(1):57–106, 1986. doi:10.24033/asens.1504.

[23] Vaughan F. R. Jones and V. S. Sunder, *Introduction to Subfactors*, London Mathematical Society Lecture Note Series, Vol. 234, Cambridge University Press, 1997. doi:10.1017/CBO9780511566219.

[24] Vaughan F. R. Jones, *Index for subfactors*, Inventiones Mathematicae, 72(1):1–25, 1983. doi:10.1007/BF01389127.

[25] Vaughan F. R. Jones, *A polynomial invariant for knots via von Neumann algebras*, Bulletin of the American Mathematical Society, 12(1):103–111, 1985. doi:10.1090/S0273-0979-1985-15304-2.

[26] H. N. V. Temperley and E. H. Lieb, *Relations between the "percolation" and "colouring" problem and other graph-theoretical problems associated with regular planar lattices*, Proceedings of the Royal Society A, 322(1549):251–280, 1971. doi:10.1098/rspa.1971.0067.

[27] Nicolas Gisin, *Bell's inequality holds for all non-product states*, Physics Letters A, 154(5–6):201–202, 1991. doi:10.1016/0375-9601(91)90805-I.

[28] B. Hensen et al., *Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres*, Nature, 526:682–686, 2015. doi:10.1038/nature15759.

[29] B. S. Tsirelson, *Quantum generalizations of Bell's inequality*, Letters in Mathematical Physics, 4:93–100, 1980. doi:10.1007/BF00417500.

[30] Alexander Grothendieck, *Résumé de la théorie métrique des produits tensoriels topologiques*, Boletim da Sociedade de Matemática de São Paulo, 8:1–79, 1953.

[31] W. K. Wootters and W. H. Zurek, *A Single Quantum Cannot Be Cloned*, Nature, 299:802–803, 1982. doi:10.1038/299802a0.

[32] W. Forrest Stinespring, *Positive Functions on C^*-Algebras*, Proceedings of the American Mathematical Society, 6:211–216, 1955. doi:10.1090/S0002-9939-1955-0069403-4.

[33] Wolfgang Pauli, *Über das Wasserstoffspektrum vom Standpunkt der neuen Quantenmechanik*, Zeitschrift für Physik, 36(5):336–363, 1926. doi:10.1007/BF01450175.

[34] Alexander Kramida, *A Critical Compilation of Experimental Data on Spectral Lines and Energy Levels of Hydrogen, Deuterium, and Tritium*, Atomic Data and Nuclear Data Tables, 96:586–644, 2010. doi:10.1016/j.adt.2010.05.001.

[35] Takehiko Shimanouchi, *Tables of Molecular Vibrational Frequencies, Consolidated Volume I*, National Standard Reference Data Series, NSRDS-NBS 39, National Bureau of Standards, 1972. doi:10.6028/NBS.NSRDS.39.

[36] Emmy Noether, *Invariant Variation Problems*, translated by M. A. Tavel, Transport Theory and Statistical Physics, 1(3):186–207, 1971; originally published 1918. doi:10.1080/00411457108231446.

[37] Manfredo P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice-Hall, 1976.

[38] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), *Observation of Gravitational Waves from a Binary Black Hole Merger*, Physical Review Letters, 116(6):061102, 2016. doi:10.1103/PhysRevLett.116.061102.

[39] S. S. Chern and J. Simons, *Characteristic forms and geometric invariants*, Annals of Mathematics, 99(1):48–69, 1974. doi:10.2307/1971013.

[40] C. N. Yang and R. L. Mills, *Conservation of Isotopic Spin and Isotopic Gauge Invariance*, Physical Review, 96(1):191–195, 1954. doi:10.1103/PhysRev.96.191.

[41] G. Arnison et al. (UA1 Collaboration), *Experimental observation of isolated large transverse energy electrons with associated missing energy at $\sqrt{s} = 540\ GeV$*, Physics Letters B, 122(1):103–116, 1983. doi:10.1016/0370-2693(83)91177-2.

[42] G. Arnison et al. (UA1 Collaboration), *Experimental observation of lepton pairs of invariant mass around 95 GeV/c^2 at the CERN SPS collider*, Physics Letters B, 126(5):398–410, 1983. doi:10.1016/0370-2693(83)90188-0.

[43] E. Witten, *Quantum field theory and the Jones polynomial*, Commun. Math. Phys., 121(3):351–399, 1989. doi:10.1007/BF01217730.

[44] Nicolai Reshetikhin and Vladimir G. Turaev, *Invariants of 3-manifolds via link polynomials and quantum groups*, Inventiones Mathematicae, 103:547–597, 1991. doi:10.1007/BF01239527.

[45] A. Achúcarro and P. K. Townsend, *A Chern–Simons action for three-dimensional anti-de Sitter supergravity theories*, Physics Letters B, 180(1–2):89–92, 1986. doi:10.1016/0370-2693(86)90140-1.

[46] Edward Witten, $2+1$ *dimensional gravity as an exactly soluble system*, Nuclear Physics B, 311(1):46–78, 1988. doi:10.1016/0550-3213(88)90143-5.

[47] Yakir Aharonov and David Bohm, *Significance of Electromagnetic Potentials in the Quantum Theory*, Physical Review, 115(3):485–491, 1959. doi:10.1103/PhysRev.115.485.

[48] Akira Tonomura, Nobuyuki Osakabe, Tsuyoshi Matsuda, Takeshi Kawasaki, Junji Endo, Shinichiro Yano, and Hiroji Yamada, *Evidence for Aharonov–Bohm effect with magnetic field completely shielded from electron wave*, Physical Review Letters, 56(8):792–795, 1986. doi:10.1103/PhysRevLett.56.792.

[49] T. Bitter and D. Dubbers, *Manifestation of Berry's topological phase in neutron spin rotation*, Physical Review Letters, 59(3):251–254, 1987. doi:10.1103/PhysRevLett.59.251.

[50] J. Nakamura, S. Liang, G. C. Gardner, and M. J. Manfra, *Direct observation of anyonic braiding statistics*, Nature Physics, 16:931–936, 2020. doi:10.1038/s41567-020-1019-1.